Analysis and Prediction of Student Academic Performance Using Data Science Techniques

1 Introduction

Education is significant in the future of people and societies. The role of factors that determine the academic performance of the students has continued to gain importance among the teachers, policymakers and also among the researchers. Now that data science and machine learning tools have evolved, one can now use educational data to extract patterns and predict student performance with greater precision.

Among various factors affecting student performance, there are such factors as family background, habits of the lifestyle, time study, and social behavior. These relationships can be identified to assist institutions in formulating superior intervention strategies that can enhance academic success.

1.1 Problem Formulation

Conventional student studies tend to examine student marks, without the part effects of behavioral and demographic variables. To this project, a data approach is adopted to:

1. Examine the demographic and lifestyle factors effect on school performance.
2. Predict end student grades based on machine learning models.

Research Question

Do you think study habits, attendance and past grades are predictive of student academic performance?

1.2 Data Acquisition

Origin: Publicly available dataset from the UCI Machine Learning Repository.

Content: Student achievement data from two Portuguese secondary schools, including demographic, social, academic, and school-related information.

Variables: Multiple columns including study time, absences, past failures, family support, internet access, first period grade (G1), second period grade (G2), and final grade (G3).

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	6	5	6	6
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	4	5	5	6
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	10	7	8	10
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	2	15	14	15
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	4	6	10	10

Figure 1: Preview of Student Performance Dataset

2 Data Analysis

2.1 Clean and preprocess data

1. **Load Dataset:** Pandas were used to import the dataset to be analyzed.
2. **Check Missing Values:** Missing values were determined and taken accordingly.
3. **Remove Duplicates:** Student records were cross-examined and eliminated where there were duplicates.
4. **Encode Categorical Variables:** Categorical variables were decided into encodes (school, gender, job type, etc.) and modified into numerical runs to be used.
5. **Data Standardization:** Numerical variables were standardized to enhance the work of the model.
6. **Feature Selection:** The selection of relevant academic and behavioral variables was chosen to be analyzed.

After cleaning, the dataset contained 395 unique rows and 42 columns.

2.2 Exploratory Data Analysis

1. Final Grade Distribution Analysis

4The histogram of final grades reveals the performance of the students to be moderately centered at the mid-range values. The average academic performance of students shown on the dataset before most students scored between 8 and 15. A few students secured very low or very high marks, implying inconsistency in the results of the learning.

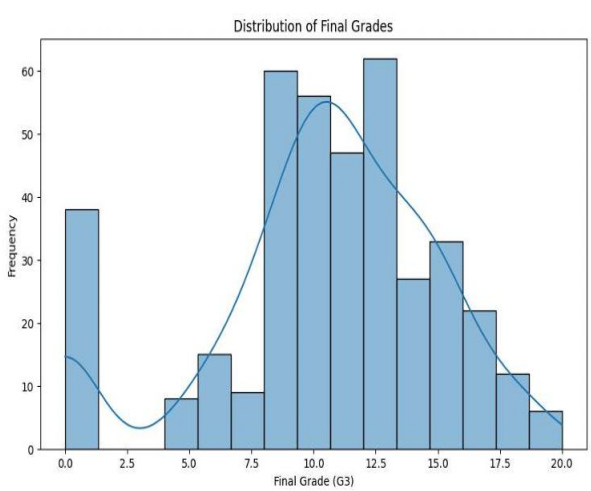


Figure 2: Distribution of Final Grades

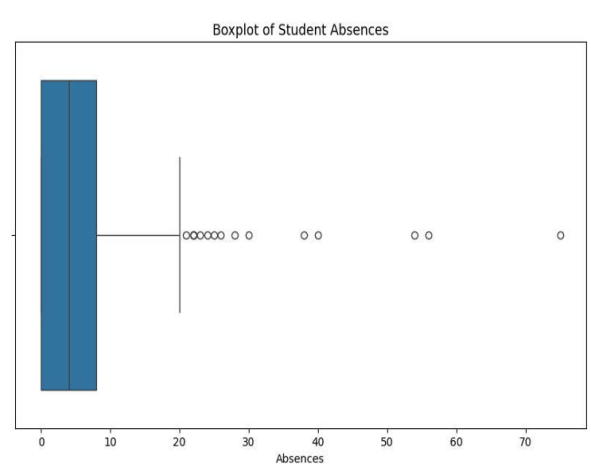


Figure 3: Boxplot of Student Absences

Also, the boxplot of the student absences reveals the existence of several outliers. Though the overall rate of more than 15 absences was lower among most students, there are some students who demonstrated high levels of absence that can adversely affect their performance.

2. Academic Performance by Study Time

The bar chart on average final grades in relation to the level of study time indicates the presence of a positive relationship between study time and academic performance. It was observed that the average final grades of students who said they studied more, were better.

This creates an implication that committed action towards academic activities leads to better student performance.

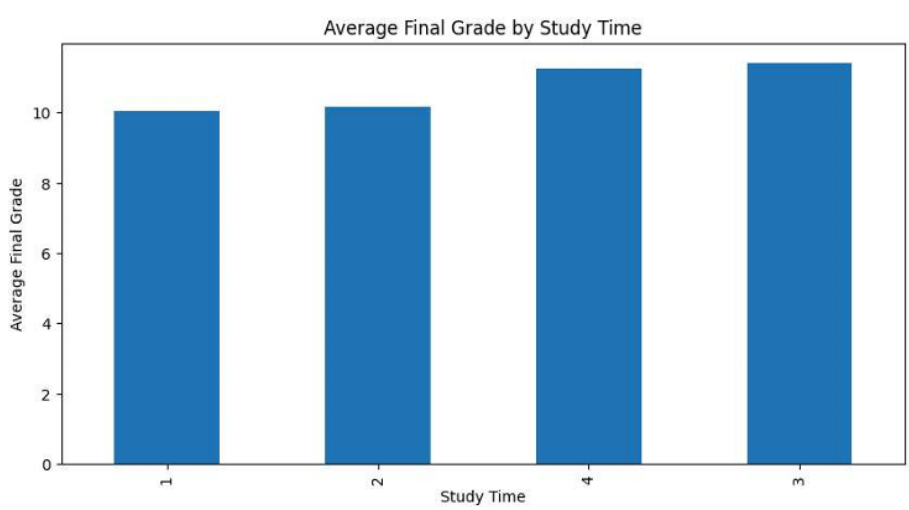


Figure 4: Average Final Grade by Study Time

3. Principal Component Analysis (PCA)

The Principal Component Analysis was used to shrink the truth values of the data and determine the key factors of student performance.

According to the scree plot, first few major components represent a significant percentage of total variance in the set of data. Following the sixth component, there is minimal contribution to the remainder of the components, and it is possible that most of the significant academic information has been covered in the initial components.

This proves that effectiveness of a student performance in terms of reduction to a smaller number of derived features is possible.

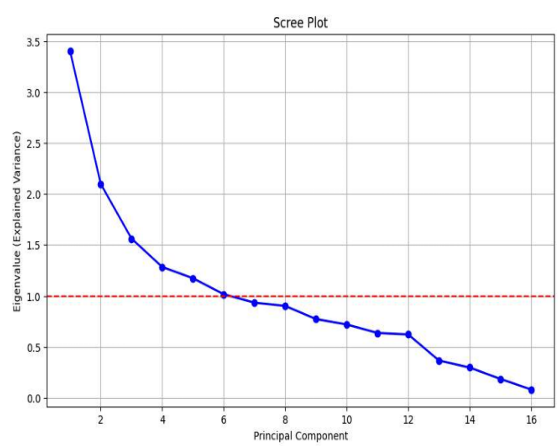


Figure 5: Scree Plot

4. Student Distribution by School

The distribution graph indicates that there is an imbalance in the data set within the schools. There are many more students enrolled in GP school and relatively fewer students in MS school.

This skew must be viewed considering the model outcomes interpretation since it can affect the predictive quality of the machine learning algorithms.

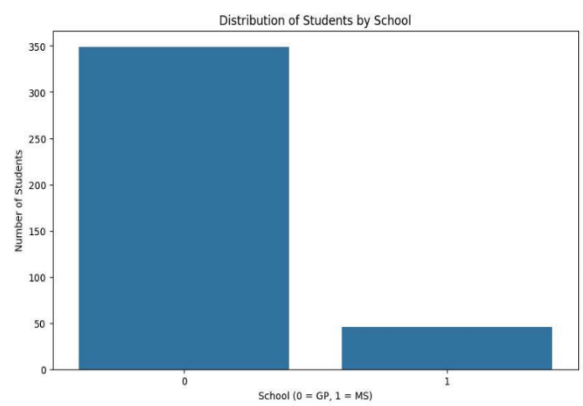


Figure 6: Distribution of Students by School

5. Comparison of Grades Across Study Time Levels

The group bar chart of grades G1, G2 and G3 at different levels of study time indicates that the performance improved throughout with the increase in amount of study time.

Students who have high levels of academic time are more likely to score higher in all the three grading periods, which supports the significance of long-term timing in academic activity.

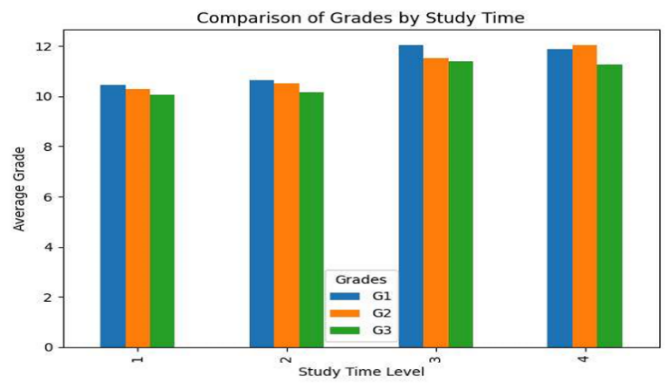


Figure 7: Comparison of Grades by Study Time

6. Correlation Analysis

The heatmap demonstrates the connections between significant academic variables.

- There is a strong positive relationship between G1, G2, and G3, which means that initial performance in academics significantly is a predictor of end grades.
- The correlation between failures and grades is shown to be moderate and negative or fails to involve much inverse correlation therefore places the truth that students who had more fails in the past yielded lower end grades.

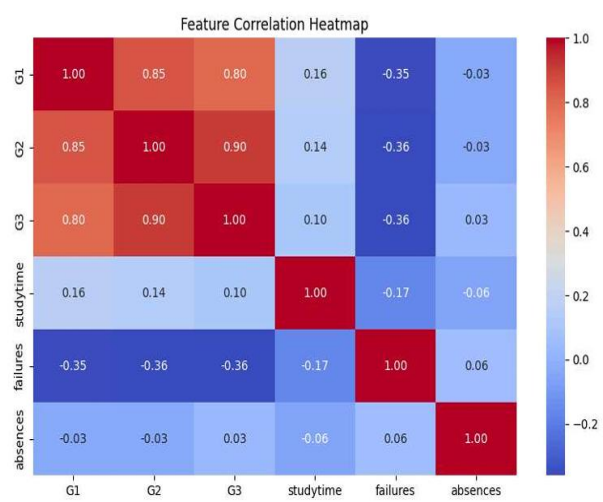


Figure 8: Feature Correlation Heatmap

- Time studying is positively related to grades mildly, whereas absences are weakly correlated.

These relationships offer useful information about academic success determinants.

These results can be further supported by scatter plots that demonstrate a linear correlation between past grades and final performance with the relationship between absences and final grades being weak.

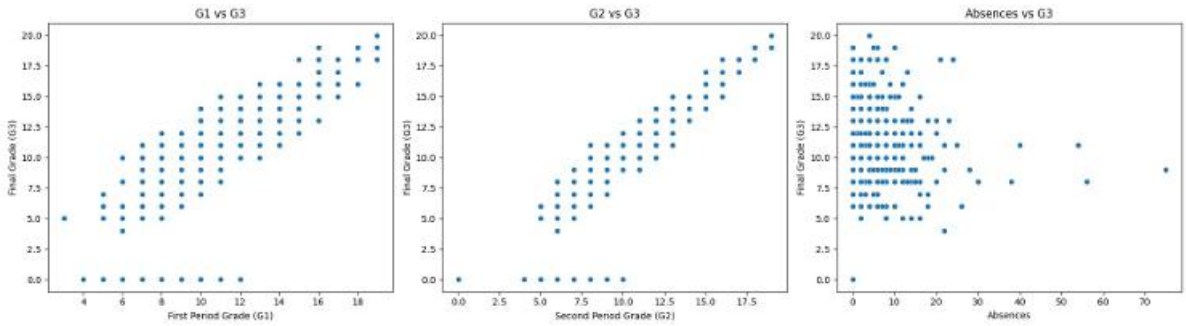


Figure 9: Scatter Plot Relationships

7. Impact of Past Failures on Academic Performance

The breakdown of average final grade as a function of failures in the past shows adverse downward trend. The students who had no prior failures scored the highest average end grades and those who had more than one failure scored much less.

This shows the implication of academic failure on the performance of a student over the long-term.

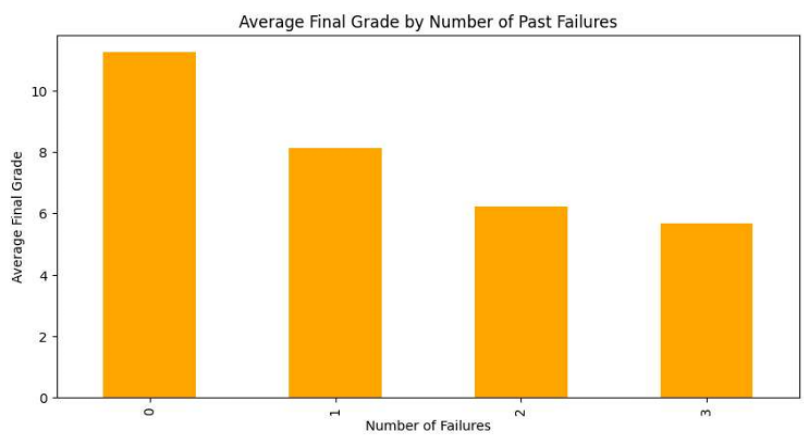


Figure 10: Average Final Grade by Number of Past Failures

8. Feature Importance Analysis

The graph of the random forest feature importance reveals that the second period grade (G2) exhibits the strongest impact on final performance. Absences, study time and first period grade

are other significant features. Such support-related variables as internet access and the support of schools are less significant in prediction accuracy.

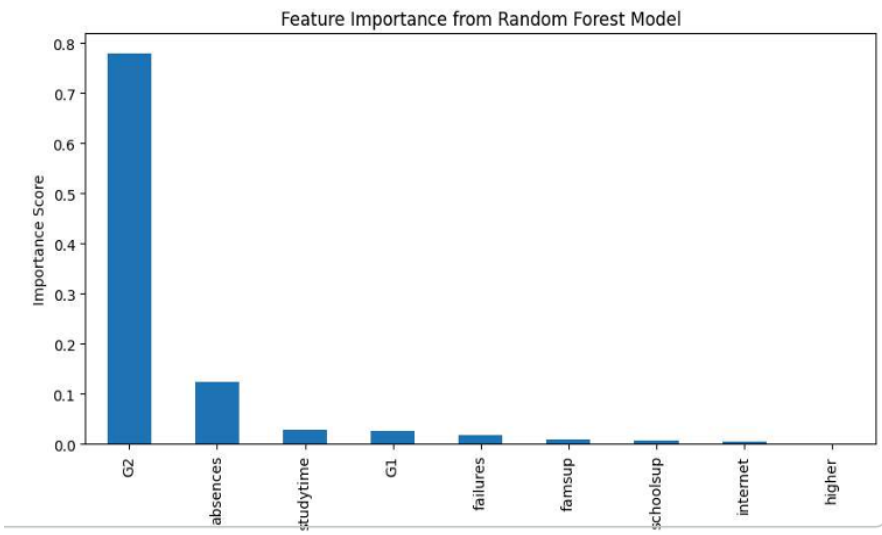


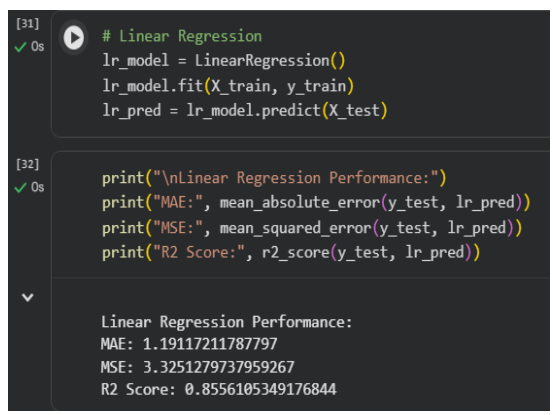
Figure 11: Feature Importance from Random Forest Model

3 Machine Learning / AI Methods

This research proceeded to use three monitored machine learning models to foresee the ultimate performance of the students (G3). The models that have been selected are Linear Regression, Random Forest Regressor, and an Artificial Neural Network. The dataset was split into training and testing part based on the 70:30 split ratio. Mean Absolute Error (MAE), Mean Squared Error (MSE) and R^2 Score was used as the measures of model performance.

3.1 Linear Regression

The Linear Regression model was applied as a control model to examine the linear association between the attributes of students and the end academic achievement. The model gave a MAE of 1.19, MSE of 3.32, and an R^2 of 0.856 which implied that the model had the ability to explain a proportion of 85.6 of the variation in the end-of-year grades of the students. The model had a good predictive insight but poor in representing intricate nonlinear relationships in the data.



```
[31]
✓ 0s
# Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)
lr_pred = lr_model.predict(X_test)

[32]
✓ 0s
print("\nLinear Regression Performance:")
print("MAE:", mean_absolute_error(y_test, lr_pred))
print("MSE:", mean_squared_error(y_test, lr_pred))
print("R2 Score:", r2_score(y_test, lr_pred))

Linear Regression Performance:
MAE: 1.19117211787797
MSE: 3.3251279737959267
R2 Score: 0.8556105349176844
```

Figure 12: Linear Regression Model

3.2 Random Forest Regressor

A Random Forest Regressor was implemented to enhance the accuracy of prediction. This final ensemble model gave more advantageous results of an MAE, MSE and an R2 of 0.93, 2.22 and 0.903. The analysis of importance of features indicated that the previous academic performance, especially the second period grade (G2) bore the greatest effect on the conclusive grade. In general, the best predictor of all the models was the Random Forest.

```
[33]
✓ 0s
# Random Forest Regressor
rf_model = RandomForestRegressor(n_estimators=100, random_state=88)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)

[34]
✓ 0s
print("\nRandom Forest Performance:")

rf_mae = mean_absolute_error(y_test, rf_pred)
rf_mse = mean_squared_error(y_test, rf_pred)
rf_r2 = r2_score(y_test, rf_pred)

print("MAE:", rf_mae)
print("MSE:", rf_mse)
print("R2 Score:", rf_r2)

Random Forest Performance:
MAE: 0.9310074029611844
MSE: 2.2240278919925114
R2 Score: 0.9034244094712677
```

Figure 13: Random Forest Regressor Model

3.3 Neural Network Model

Multiple dense layers and dropout regularization were also used to create a feed-forward Neural Network in order to alleviate overfitting. It was trained in 100 epochs with the Adam optimizer and Mean Squared error loss function. The analysis findings indicated an MAE of 1.78, the MSE of 4.79, and an R^2 value of 0.796. Even though the neural network had the capability to learn the patterns in the data, its performance was a bit lower compared to the Random Forest model since the dataset was structured and large.

```
[36]
✓ 12s
# Neural Network Model
nn_model = Sequential([
    Dense(128, activation='relu', input_shape=(X_train.shape[1],)),
    Dropout(0.2),
    Dense(64, activation='relu'),
    Dropout(0.2),
    Dense(32, activation='relu'),
    Dense(1)   # Output layer for regression
])

# Compile the model
nn_model.compile(optimizer='adam', loss='mse', metrics=['mae'])

# Train the model
history = nn_model.fit(
    X_train, y_train,
    epochs=100,
    batch_size=16,
    validation_data=(X_test, y_test),
    verbose=1
)
```

Figure 14: Neural Network Model

3.4 Model Comparison

The comparison of the performances of the models shows that the highest accuracy of the predictions is obtained via the Random Forest Regressor model, and then the Linear Regression and Neural Network model follow.

Model	MAE	MSE	R^2
Linear Regression	1.19	3.32	0.856
Random Forest	**0.93**	**2.22**	**0.903**
Neural Network	1.78	4.79	0.796

Table 1: Model Comparison Table

Consequently, the Random Forest model was selected as the best model on predicting academic performance of students in this analysis.

4 Reflection

4.1 Challenges Faced

Missing Values:

In the stage of pre-processing data, some of the students' records had extreme values in terms of absence and failure in the past. These extreme values had an impact on the distribution of variables and affected the performance of the models. The most significant issue was the question of keeping these values or modifying them without highlighting the useful academic behavioral patterns.

The Relationship of features or Multicollinearity:

A high degree of correlation was found between the past grades of the students (G1 and G2) and the final grade (G3). Such multicollinearity not only complicated the interpretation of the one-on-one effect of each predictor in regression models but also necessitated the selection of features.

4.2 Key Learnings

Significance of the Past Academic Performance:

The examination showed that previous scores were the most effective predictors of the ultimate academic performance of the students. This creates the cumulative nature of learning outcomes within learning settings.

Performance of Dimensionality Reduction:

Per the Principal Component Analysis (PCA), a comparatively small number of components could describe much of the variability in the data. This contributed towards the explanation of the internal structure of the factors of student performance.

Comparison Summary Model Performance:

The Random Forest Regressor was found to have the largest predictive accuracy compared to the Linear Regression and the Neural Network model. Ensemble learning methods were more useful in the representation of nonlinear relationships in data of students.

Metric	Linear Regression	Random Forest	Neural Network	Best Performer
R^2 Score	0.856	**0.903**	0.796	Random Forest
MAE	1.19	**0.93**	1.78	Random Forest
MSE	3.32	**2.22**	4.79	Random Forest

Table 2: Model Performance Comparison

To enhance my learning of machine learning principles and analytical methods, I consulted scholarly materials, Internet tutorials and experimentation with Python packages, Scikit-learn and TensorFlow. This project broadened my understanding on how to preprocess, visualize, evaluate and compare models in predictive modelling.

5 References

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